

1. A 42-year-old woman, gravida 2, para 2, comes to the physician requesting contraceptive advice. She is not married, but she has been with the same partner for 8 years, and she has decided that her family is complete. She has smoked one pack of cigarettes daily for 20 years. She is 165 cm (5 ft 5 in) tall and weighs 59 kg (130 lb); BMI is 22 kg/m². Her blood pressure is 130/80 mm Hg. Examination shows no abnormalities. Which of the following contraceptive methods is contraindicated for this couple?
- ☐ A) Diaphragm
 - ☐ B) IUD
 - ☐ C) Oral contraceptives
 - ☐ D) Spermicidal foam
 - ☐ E) Tubal ligation
 - ☐ F) Vasectomy

2. A 38-year-old woman, gravida 2, para 1, at 26 weeks' gestation comes to the physician for a routine prenatal visit. At 11 weeks' gestation, she underwent chorionic villus sampling, which showed a 46,XX karyotype. Her first child has Down syndrome. The patient's immunizations are up-to-date. Her only medication is a prenatal vitamin. Her pregnancy has been complicated by asymptomatic bacteriuria with *Streptococcus agalactiae* (group B), which was treated at 12 weeks' gestation with ampicillin. Examination shows a uterus consistent in size with a 26-week gestation. The maternal blood group is B, Rh-negative. Serum rubella antibody titer is nonimmune. Repeat urine culture is negative. Which of the following is the most appropriate next step in management?
- ☐ A) Administration of measles-mumps-rubella vaccine at this visit
 - ☐ B) Administration of Rh₀(D) immune globulin if antibody screening is negative at this visit
 - ☐ C) Anovaginal culture for *Streptococcus agalactiae* (group B) at 36 weeks' gestation
 - ☐ D) Fluorescent treponemal antibody assay at 28 weeks' gestation
 - ☐ E) Repeat urine culture now and during the third trimester

3. An 82-year-old woman comes to the physician because of a 3-month history of urinary urgency; during this period, she has noticed that the toilet paper is blood-tinged after urinating. At times she loses small amounts of urine unrelated to any precipitating events. She has a strong urge to urinate almost all the time. She voids small amounts every 30 minutes during the day and awakens approximately two times nightly to urinate. A photograph of the genitalia is shown. The remainder of the examination shows no abnormalities. Urinalysis shows no RBCs, WBCs, bacteria, or protein. A Gram stain of urine shows no bacteria. Which of the following is the most likely diagnosis?

- ☐ A) Urethral carcinoma
- ☐ B) Urethral diverticulum
- ☐ C) Urethritis secondary to estrogen deficiency
- ☐ D) Urethrocele
- ☐ E) Urethrovaginal fistula



4. Two days after undergoing cesarean delivery because of failure to progress, a 27-year-old woman, gravida 1, para 1, has mild chest pain and dyspnea after walking to the nursery to see her newborn. Her pregnancy had been uncomplicated. She has no history of serious illness and takes no medications. Her temperature is 38.1°C (100.6°F), pulse is 110/min, respirations are 28/min, and blood pressure is 140/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 90%. Physical examination shows a clean, dry incision. Cardiopulmonary examination shows no abnormalities. The abdomen is moderately tender. Pelvic examination shows moderate lochia. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Abdominal ultrasonography
 - ☐ B) Doppler ultrasonography of the lower extremities
 - ☐ C) Echocardiography
 - ☐ D) Spiral CT scan of the chest
 - ☐ E) MRI of the abdomen



5. A 27-year-old woman, gravida 2, para 1, at 16 weeks' gestation comes to the physician because of moderate pain with urination during the past 2 days. She has no history of serious illness, and pregnancy had been uncomplicated. Examination shows a uterus consistent in size with a 16-week gestation. Urine culture grows greater than 100,000 colonies/mL of group B streptococcus, sensitive to penicillin. In addition to antibiotic therapy now, which of the following is the most appropriate management?
- ☐ A) Screening for group B streptococcal infection at 35 weeks' gestation
 - ☐ B) Screening for group B streptococcal infection at the onset of labor
 - ☐ C) Maternal antibiotic therapy at 35 weeks' gestation
 - ☐ D) Maternal antibiotic therapy during labor
 - ☐ E) Elective cesarean delivery at 38 weeks' gestation



6. A 33-year-old nulligravid woman comes to the physician for a premarital examination. She has had increased cramping with menses over the past 6 months. Menses occur at regular intervals with occasional spotting 3 days before onset. She has had occasional dyspareunia over the past 8 months. Pelvic examination shows normal external genitalia, vagina, and cervix. The uterus is firm, anterior, and normal-sized. The ovaries are palpable bilaterally and are normal-sized. Two 1-cm tender nodules are palpated on the right uterosacral ligament. This woman is at increased risk for which of the following conditions?
- ☐ A) Endometrial cancer
 - ☐ B) Endometritis
 - ☐ C) Infertility
 - ☐ D) Leiomyosarcoma
 - ☐ E) Recurrent abortion



7. A 52-year-old woman comes to the physician because of a 3-month history of severe left groin pain. Initially, the pain occurred only with activity, but it is now constant and awakens her at night. She does not have pain in any other location. During the past 6 weeks, she has had a decreased appetite resulting in a 4.5-kg (10-lb) weight loss. She has no history of injury to the groin area. Her only medication is ibuprofen. Range of motion of the left hip elicits pain. Sensation and motor function are intact. Distal pulses are 1+. X-rays of the pelvis and left hip show a poorly defined, mixed, lytic and blastic lesion in the femoral neck and intertrochanteric region. A whole-body bone scan shows areas of uptake in the ribs, spine, and left proximal femur. Which of the following is the most likely site of origin of this lesion?
- ☐ A) Breast
 - ☐ B) Colon
 - ☐ C) Endometrium
 - ☐ D) Ovary
 - ☐ E) Stomach



8. A 27-year-old woman comes to the office because of a 1-week history of malodorous vaginal discharge. She has not had abdominal pain, pain with urination, or fever. Her last menstrual period was 1 week ago. Menses occur at regular 28-day intervals and last 3 days; flow is normal. Three months ago, she had vulvovaginal candidiasis that resolved after treatment with clotrimazole. Her only current medication is a combination oral contraceptive. She is sexually active with two male partners; they use condoms inconsistently. Her temperature is 37°C (98.6°F), pulse is 72/min, and blood pressure is 120/82 mm Hg. Abdominal examination shows no abnormalities. Pelvic examination shows a moderate amount of homogeneous, gray-white discharge in the vagina; the vagina otherwise appears normal. The cervix appears normal. There is no cervical motion, uterine, or adnexal tenderness. The adnexa are not enlarged. Wet mount preparation of the vaginal discharge is performed. Addition of KOH produces a fishy odor. Microscopic examination shows clue cells; no other abnormalities are noted. Which of the following is the most appropriate next step in management?
- ☐ A) Diluted vinegar douche
 - ☐ B) Intramuscular ceftriaxone and oral doxycycline for the patient and oral doxycycline for her sexual partners
 - ☐ C) Oral fluconazole for the patient only
 - ☐ D) Oral metronidazole for the patient and her sexual partners
 - ☐ E) Intravaginal metronidazole for the patient



9. A 32-year-old woman, gravida 5, para 4, at 33 weeks' gestation is admitted to the hospital because of a 2-hour history of vaginal bleeding. She has not had pain, fever, or contractions. She had sexual intercourse 2 days ago. She has received no prenatal care. Her first pregnancy ended in an uncomplicated spontaneous vaginal delivery at term. Her other pregnancies ended in elective repeat cesarean deliveries. She takes no medications. A Pap smear 2 years ago was reported as atypical squamous cells of undetermined significance. Human papillomavirus testing at that time was negative for high-risk types. She uses cocaine socially. She does not appear to be in distress. Her temperature is 37.3°C (99.1°F), pulse is 80/min, and blood pressure is 140/95 mm Hg. The abdomen is soft and nontender. Fundal height is 34 cm. The fetal heart rate is reassuring. No contractions are detected on tocometry. There is a small amount of dark red vaginal bleeding. Which of the following is the most likely diagnosis?
- ☐ A) Abruptio placentae
 - ☐ B) Cervical cancer
 - ☐ C) Placenta previa
 - ☐ D) Premature contractions
 - ☐ E) Vaginal laceration



10. A 62-year-old woman comes to the office because of temperatures to 38.3°C (101°F) for 5 days and an intermittent watery vaginal discharge for 2 days. The discharge is worse during the day and when standing. She underwent a vaginal repair and hysterectomy 2 weeks ago. She has no loss of urine with coughing or sneezing. Which of the following is the most likely diagnosis?
- ☐ A) Cystitis
 - ☐ B) Cystocele
 - ☐ C) Rectovaginal fistula
 - ☐ D) Urethral prolapse
 - ☐ E) Urethritis
 - ☐ F) Urethrocele
 - ☐ G) Vaginal vault prolapse
 - ☐ H) Vesicovaginal fistula



11. A 23-year-old woman comes to the physician because of temperatures to 37.8°C (100°F), malaise, multiple painful blisters on the vulva, and difficulty voiding for 36 hours. She is sexually active with multiple partners. Examination shows multiple tender vesicles on the vulva and tender inguinal lymphadenopathy. Which of the following is the most likely diagnosis?
- ☐ A) Chancroid
 - ☐ B) *Chlamydia trachomatis* infection
 - ☐ C) Herpes genitalis
 - ☐ D) Lymphogranuloma venereum
 - ☐ E) Molluscum contagiosum
 - ☐ F) Trichomoniasis



12. One day after undergoing a cesarean delivery for arrest of dilation, a 32-year-old woman, gravida 1, para 1, has a temperature of 38.1°C (100.6°F) and mild shortness of breath. She received prophylactic antibiotics at delivery. Her pulse is 110/min, respirations are 14/min, and blood pressure is 110/72 mm Hg. Breath sounds are decreased bilaterally. No rhonchi, wheezes, or crackles are heard. Examination of the breasts shows no engorgement or tenderness. Abdominal examination shows a moderately tender fundus; the surgical incision is clean and dry. Which of the following is the most likely diagnosis?
- ☐ A) Atelectasis
 - ☐ B) Bacterial pneumonia
 - ☐ C) Endomyometritis
 - ☐ D) Pelvic abscess
 - ☐ E) Septic pelvic thrombophlebitis



13. A previously healthy 32-year-old woman comes to the physician because of a 3-day history of vaginal spotting. Her last menstrual period was 5 weeks ago. She takes no medications. Vital signs are within normal limits. Pelvic examination shows blood at the cervical os and a normal-sized uterus. There is no adnexal tenderness. Her quantitative serum β -hCG concentration is 1000 mIU/mL. Urine β -hCG testing is positive. Ultrasonography shows a gestational sac. There is a simple cyst on the right ovary. Which of the following is the most likely diagnosis?
- ☐ A) Abnormal intrauterine pregnancy
 - ☐ B) Ectopic pregnancy
 - ☐ C) Hydatidiform mole
 - ☐ D) Ovarian germ cell tumor
 - ☐ E) Normal early intrauterine pregnancy



14. A 32-year-old woman comes to the office because she has not had a menstrual period for 8 months; she has had hot flashes and vaginal discomfort with coitus during this time. Her last child was born 4 years ago. Menarche occurred at the age of 12 years, and menses occurred at regular monthly intervals until the age of 25 years. From the age of 25 years until she stopped menstruating, her periods became progressively less frequent. Two years ago, oral progesterone caused her to have a menstrual period, but an identical dose failed to cause a menstrual period 3 months ago. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Endometrial biopsy
 - ☐ B) Hysteroscopy
 - ☐ C) Measurement of serum estradiol-17 β concentration
 - ☐ D) Measurement of serum follicle-stimulating hormone concentration
 - ☐ E) Measurement of serum thyroid-stimulating hormone concentration



15. Ten days after the delivery of a healthy newborn, a 21-year-old woman, gravida 1, para 1, comes to the physician because of a 1-day history of heavy, bright red, vaginal bleeding. She also has had cramping abdominal pain. Her temperature is 37°C (98.6°F). Examination shows a firm uterus consistent in size with an 8- to 10-week gestation. There is blood at the cervical os. Which of the following is the most likely diagnosis?
- ☐ A) Cervical laceration
 - ☐ B) Endometritis
 - ☐ C) Retained placental fragments
 - ☐ D) Uterine atony
 - ☐ E) Uterine inversion



16. A healthy 32-year-old woman, gravida 3, para 2, at 24 weeks' gestation comes to the physician in September requesting an influenza virus immunization. She does not smoke. She is a nursing assistant at a residential facility for disabled adults. Pregnancy has been uncomplicated. Examination shows a uterus consistent in size with a 24-week gestation. The fetal heart rate is 120/min. Which of the following is the most appropriate recommendation for this patient?
- ☐ A) Antiviral prophylaxis only when cases of influenza occur in the residential facility
 - ☐ B) Influenza virus immunization after delivery
 - ☐ C) Influenza virus immunization after the influenza season begins
 - ☐ D) Influenza virus immunization only when cases of influenza occur in the residential facility
 - ☐ E) Influenza virus immunization prior to the influenza season



17. A 24-year-old nulligravid woman comes to the physician because she has been unable to conceive for 2 years. Menses occur at regular 28-day intervals, and she has mild pain with menstrual periods. She has not had pain during sexual intercourse or discharge from her breasts. At the age of 18 years, she was hospitalized for management of pelvic inflammatory disease with intravenous antibiotic therapy. At that time, nucleic acid amplification testing of the cervix for *Chlamydia trachomatis* was positive, and testing for *Neisseria gonorrhoeae* was negative. Her husband has a 2-year-old child from a previous relationship. Physical examination of the patient, including pelvic examination, shows no abnormalities. Hysterosalpingography shows bilateral hydrosalpinx and distal tubal occlusion. Which of the following is the most likely cause of the imaging findings?
- ☐ A) Congenital defect
 - ☐ B) Ectopic endometrial tissue
 - ☐ C) Immune response
 - ☐ D) Ovarian cancer
 - ☐ E) Tubal spasm



18. A 32-year-old primigravid woman at 7 weeks' gestation is admitted to the hospital because of a 2-day history of nausea, vomiting, and dehydration. She has had a 4-week history of generalized weakness. She has been admitted twice during the past 2 weeks for similar symptoms. She has no history of gastrointestinal illness. Her only medication is a prenatal vitamin. The patient is in mild distress. Her temperature is 37°C (98.6°F), pulse is 125/min, and blood pressure is 120/80 mm Hg. Ophthalmologic examination shows mild exophthalmos bilaterally. There is a resting tremor of the hands. Measurement of which of the following serum concentrations is the most appropriate next step in diagnosis?
- ☐ A) Free thyroxine
 - ☐ B) Free triiodothyronine
 - ☐ C) Thyroxine-binding globulin
 - ☐ D) Total thyroxine
 - ☐ E) Total triiodothyronine



19. A 27-year-old primigravid woman at 39 weeks' gestation is admitted to the hospital in labor. Contractions began 6 hours ago and are occurring every 3 minutes. Her pregnancy has been uncomplicated. The cervix is 6 cm dilated and 100% effaced; the vertex is at -1 station. The membranes have not ruptured. Two hours later, the cervical examination is unchanged despite adequate, regular contractions at 3-minute intervals. Which of the following is the most likely cause of these findings?
- ☐ A) Arrest of descent
 - ☐ B) Cervical stenosis
 - ☐ C) Contracted pelvis
 - ☐ D) Prolonged active phase
 - ☐ E) Prolonged latent phase



20. A 23-year-old woman comes to the physician because of a 4-day history of bloody discharge from her left nipple. Her maternal grandmother was diagnosed with breast cancer at the age of 37 years. Breast examination shows bloody discharge from the left nipple and a 2-mm nodule underneath the left areola. Which of the following is the most likely diagnosis?

- ☐ A) Breast abscess
- ☐ B) Breast cancer
- ☐ C) Cystosarcoma phyllodes
- ☐ D) Ductal papilloma
- ☐ E) Fat necrosis of the breast
- ☐ F) Fibroadenoma
- ☐ G) Fibrocystic changes of the breast
- ☐ H) Galactocele
- ☐ I) Inflammatory carcinoma
- ☐ J) Mastitis

21. A 27-year-old woman, gravida 2, para 1, at 24 weeks' gestation comes to the physician for a routine prenatal visit. Two weeks ago, she fell off her bicycle and had no visible injuries, so she did not seek treatment. Her pregnancy has been otherwise uncomplicated. The patient's blood group is A, Rh positive. Her only medications are a prenatal vitamin and iron supplementation. Vital signs are within normal limits. Examination shows a fundal height of 28 cm. Ultrasonography shows polyhydramnios and no other abnormalities. Which of the following is the most likely cause of these findings?
- ☐ A) Fetal esophageal atresia
 - ☐ B) Fetal growth restriction
 - ☐ C) Fetal renal agenesis
 - ☐ D) Fetal Rh alloimmunization
 - ☐ E) Maternal trauma



22. A 32-year-old woman with hirsutism and oligomenorrhea comes to the physician for a follow-up examination 6 months after beginning prednisolone therapy. Six months ago, her serum 17 α -hydroxyprogesterone concentration was 850 ng/dL (N=20–300). Within 2 months, menses began occurring at regular 28-day intervals. Examination today shows a decrease in hirsutism. A second measurement of serum 17 α -hydroxyprogesterone concentration is 195 ng/dL. Which of the following is the most likely explanation for improvement in this patient's symptoms?
- ☐ A) Increased aromatase activity
 - ☐ B) Increased hepatic synthesis of sex hormone-binding globulin
 - ☐ C) Increased renal clearance of androgens
 - ☐ D) Ovarian androgen synthesis
 - ☐ E) Suppressed adrenal androgen synthesis

23. A 27-year-old woman comes to the physician because of a 2-day history of pain in her right knee and left wrist and spots on her extremities. Three nights ago, she ate pizza at a local restaurant with some friends, but they did not get sick. Her last menstrual period began 3 days ago, and she is currently using tampons. She has had unprotected sexual intercourse with two asymptomatic partners over the past month and uses an oral contraceptive. She visited Cape Cod 4 months ago but does not recall any illness or symptoms at that time. Her temperature is 39°C (102.2°F). Examination shows scattered pustular lesions on the extremities including one on the left palm and one on the right great toe. The left wrist is slightly red, and the redness extends along the extensor tendons. Which of the following is most likely to confirm the diagnosis?
- ☐ A) Antistreptolysin O titer
 - ☐ B) Serologic testing for syphilis
 - ☐ C) Testing for *Borrelia burgdorferi*
 - ☐ D) Testing for *Neisseria gonorrhoeae*
 - ☐ E) Testing for *Staphylococcus aureus*



24. A 31-year-old primigravid woman at 23 weeks' gestation is brought to the emergency department because of a 1-day history of moderate lower abdominal pain, decreased appetite, and nausea. She has no history of serious illness. Her only medication is a prenatal vitamin. Her temperature is 38°C (100.4°F), pulse is 70/min, and blood pressure is 100/90 mm Hg. Examination shows moderate tenderness to palpation of the right mid abdomen. The uterus is consistent in size with a 23-week gestation. There is no bleeding or fluid discharge. Her leukocyte count is 15,000/mm³, serum AST activity is 45 U/L, and serum ALT activity is 48 U/L. Which of the following is the most likely diagnosis?
- ☐ A) Abruptio placentae
 - ☐ B) Appendicitis
 - ☐ C) Braxton Hicks contractions
 - ☐ D) Hepatitis
 - ☐ E) Preterm labor



25. A 17-year-old girl is brought to the physician by her mother because of a 5-month history of painful menstruation that causes her to occasionally miss school. Her last menstrual period was 3 weeks ago, and her menses occur at regular 28-day intervals. During questioning with her mother present, the patient says that she has never been sexually active. However, after the mother leaves the room, the patient indicates that she is sexually active. She and her partner use condoms consistently. Physical and pelvic examinations show no abnormalities. Which of the following is the most appropriate screening test to obtain at this visit?
- ☐ A) Complete blood count
 - ☐ B) Determination of rubella status
 - ☐ C) Pregnancy test
 - ☐ D) Testing for *Chlamydia trachomatis* and *Neisseria gonorrhoeae*
 - ☐ E) Urinalysis

26. A 25-year-old primigravid woman comes to the physician for her first prenatal visit after a home pregnancy test was positive. Her last menstrual period was 8 weeks ago. She reports that her blood group is A, Rh-negative. She has no history of chickenpox or other serious illness and takes no medications. Physical examination shows an enlarged nontender uterus consistent in size with an 8-week gestation. Which of the following is the most appropriate next step in management?
- ☐ A) Follow-up examination in 4 weeks
 - ☐ B) Measurement of fasting serum glucose concentration
 - ☐ C) Administration of Rh_o(D) immune globulin
 - ☐ D) Administration of rubella vaccine
 - ☐ E) Administration of varicella vaccine



27. A 28-year-old woman, gravida 2, para 1, at 38 weeks' gestation is brought to the office because of fluid leaking from her vagina. She has no history of serious illness. Pregnancy had otherwise been uncomplicated. At her last prenatal visit, culture for *Streptococcus agalactiae* (group B) grew no organisms. Her only medication is a prenatal vitamin. She has no known drug allergies. Her temperature is 37.0°C (98.6°F), pulse is 86/min, respirations are 16/min, and blood pressure is 106/70 mm Hg. Examination shows a uterus consistent in size with a 38-week gestation. Fetal monitoring shows no fetal distress. There are no contractions. Pelvic examination shows a nondilated cervix. Microscopic examination of the fluid shows ferning. Which of the following is the most appropriate next step in management?
- ☐ A) Admission to the hospital for induction of labor
 - ☐ B) Admission to the hospital for intravenous ampicillin therapy
 - ☐ C) Admission to the hospital for intravenous clindamycin therapy
 - ☐ D) Admission to the hospital for observation only
 - ☐ E) Follow-up office examination in 1 day

28. A 42-year-old nulligravid woman comes to the emergency department because of a 3-week history of heavy vaginal bleeding. Menarche was at the age of 12 years. Menses have occurred twice yearly during the past 10 years. Menses previously occurred at regular 28-day intervals. She has chronic hypertension treated with a calcium-channel blocking agent. She has smoked one-half pack of cigarettes daily for 15 years. She is 173 cm (5 ft 8 in) tall and weighs 72 kg (158 lb); BMI is 24 kg/m². Her blood pressure is 130/80 mm Hg. Pelvic examination shows no abnormalities. Pelvic ultrasonography shows a thickened endometrial stripe and a 2 × 2-cm, simple left adnexal cyst. Results of an endometrial biopsy show well-differentiated endometrial adenocarcinoma. Which of the following is the strongest predisposing risk factor for this patient's condition?
- ☐ A) Age
 - ☐ B) BMI
 - ☐ C) Cigarette smoking
 - ☐ D) Hypertension
 - ☐ E) Menstrual history

29. A 27-year-old woman, gravida 1, para 1, comes to the physician 2 weeks after noticing a mass in her right breast. Her child was born 1 year ago following an uncomplicated pregnancy and delivery; she discontinued breast-feeding 4 weeks ago. She has no history of serious illness and has had no operative procedures. Vital signs are within normal limits. Examination of the right breast shows a 3-cm, slightly tender, nonerythematous, mobile, fluctuant mass. The remainder of the physical examination shows no abnormalities. Fine-needle aspiration of the mass yields 5 mL of clear fluid, and the mass completely resolves. Which of the following is the most likely diagnosis?
- ☐ A) Breast abscess
 - ☐ B) Fibroadenoma
 - ☐ C) Galactocele
 - ☐ D) Inflammatory carcinoma
 - ☐ E) Mastitis

30. A 37-year-old woman at 36 weeks' gestation is brought to the emergency department 30 minutes after the sudden onset of abdominal cramping and vaginal bleeding. Two hours later, she becomes obtunded. Her temperature is 36.7°C (98.1°F), pulse is 120/min, respirations are 16/min and shallow, and blood pressure is 110/60 mm Hg. Examination shows a diffuse petechial rash. There is bleeding from the venipuncture and intravenous catheter sites. Laboratory studies show:

Hemoglobin	10 g/dL
Leukocyte count	11,000/mm ³
Platelet count	80,000/mm ³
Prothrombin time	18 sec (INR=1.5)
Partial thromboplastin time	40 sec
Plasma fibrinogen	50 mg/dL (N=200–400)
Fibrin degradation products	40 µg/mL (N<10)

Which of the following should be administered prior to delivery?

- ☐ A) Fresh frozen plasma
- ☐ B) Heparin
- ☐ C) Oxytocin
- ☐ D) Packed red blood cells
- ☐ E) Platelets

31. A 15-year-old primigravid patient at 34 weeks' gestation is brought to the emergency department because of a severe headache and blurred vision for 6 hours. Pregnancy had been uncomplicated. At her last visit 1 week ago, her blood pressure was 110/68 mm Hg, and urinalysis showed no protein. Her blood pressure today is 180/112 mm Hg. During the examination, the patient has a generalized tonic-clonic seizure. During the seizure, the fetal heart rate decreases to 60/min for 90 seconds before returning to a baseline of 140/min with good variability. Urinalysis now shows 2+ protein. Which of the following is the most appropriate next step in management?
- ☐ A) ECG
 - ☐ B) CT scan of the head
 - ☐ C) Administration of magnesium sulfate
 - ☐ D) Lumbar puncture
 - ☐ E) Immediate cesarean delivery



32. A 16-year-old girl comes to the physician because of a 2-year history of cramps, nausea, vomiting, and diarrhea for the first 2 days of menses. Menarche occurred at the age of 12 years. Menses occur at regular 28- to 30-day intervals. She is sexually active with one partner. She and her partner usually use condoms for contraception. Pelvic examination shows no abnormalities. Which of the following is the most likely explanation for these findings?
- ☐ A) Adenomyosis of the uterus
 - ☐ B) Chronic pelvic inflammatory disease
 - ☐ C) Endometriosis of the pelvic peritoneum
 - ☐ D) Increased prostaglandin production by the endometrium
 - ☐ E) Submucous leiomyomata uteri



33. Eight days after spontaneous vaginal delivery, a 32-year-old woman, gravida 1, para 1, comes to the physician because of a 5-day history of temperatures to 38°C (100.4°F) and perineal pain. The pain has increased despite treatment with narcotic analgesics and sitz baths. During delivery, she had an uncomplicated midline episiotomy repair that was closed in layers with absorbable sutures. Several bleeding veins were separately ligated using chromic sutures to control the bleeding. Her temperature today is 38.4°C (101.1°F). Examination shows tenderness and erythema over the midline adjacent to the perineal incision. The sutures are still in place. The perineum is warm. There are several large, soft hemorrhoids. Digital rectal examination produces severe discomfort; no fluctuation is palpated anteriorly or posteriorly. Which of the following is the most likely diagnosis?
- ☐ A) Infected episiotomy
 - ☐ B) Intersphincteric anal abscess
 - ☐ C) Ischiorectal fossa abscess
 - ☐ D) Perirectal abscess
 - ☐ E) Rectovaginal fistula



34. A 32-year-old woman, gravida 3, para 2, at 16 weeks' gestation comes to the physician for a routine prenatal visit. Pregnancy has been uncomplicated. Her previous pregnancies ended in spontaneous vaginal deliveries at 32 and 34 weeks' gestation. She has no history of serious illness, and her only medication is a prenatal vitamin. Examination shows a uterus consistent in size with a 16-week gestation. The fetal heart rate is 140/min. Which of the following is the most appropriate next step in management?
- ☐ A) Bed rest
 - ☐ B) Cervical cerclage
 - ☐ C) 17 α -Hydroxyprogesterone therapy
 - ☐ D) Nifedipine therapy
 - ☐ E) Terbutaline therapy



35. A 22-year-old woman, gravida 3, para 2, at 34 weeks' gestation comes to the physician for a routine prenatal visit. She immigrated to the United States from Zimbabwe 6 months ago. Pregnancy has been uncomplicated with no vaginal bleeding, contractions, or premature rupture of membranes. Examination today shows a fundal height of 30 cm. Ultrasonography shows microcephaly. Which of the following is most likely to have prevented these findings?
- ☐ A) Administration of antibiotics during pregnancy
 - ☐ B) Folic acid supplementation during pregnancy
 - ☐ C) Iron supplementation during pregnancy
 - ☐ D) Measles-mumps-rubella vaccine prior to conception
 - ☐ E) Progesterone supplementation prior to conception



36. A previously healthy 47-year-old woman, gravida 2, para 2, comes to the physician because of a 4-hour history of heavy vaginal bleeding following sexual intercourse. She has soaked one sanitary pad during this time. Her last menstrual period was 21 days ago. She became sexually active at the age of 16 years. She has smoked two packs of cigarettes daily for 33 years. Pelvic examination shows a small friable lesion on the cervix. A biopsy specimen of the lesion shows carcinoma in situ. Cone biopsy specimen of the cervix shows microinvasive squamous cell carcinoma. Which of the following is the most appropriate next step in management?
- ☐ A) Colposcopic laser ablation
 - ☐ B) Endometrial ablation
 - ☐ C) Hysteroscopic excision
 - ☐ D) Laparoscopic excision
 - ☐ E) Cryotherapy
 - ☐ F) Uterine artery embolization
 - ☐ G) Hysterectomy
 - ☐ H) Myomectomy

37. A 21-year-old primigravid woman at 34 weeks' gestation comes to the physician for a routine prenatal visit. She has hyperthyroidism treated with propylthiouracil and tachycardia controlled with labetalol. Her pulse is 90/min, and blood pressure is 120/70 mm Hg. The fetal heart rate is 140/min. Examination shows no abnormalities. Ultrasonography shows a fetus consistent in size with gestational age and a mass in the fetal neck. The estimated fetal weight is at the 15th percentile. The amniotic fluid index is 27 cm (N=10–20). Which of the following is the most likely cause of these fetal findings?
- ☐ A) Cystic hygroma
 - ☐ B) Fetal goiter
 - ☐ C) Fetal growth restriction
 - ☐ D) Gestational diabetes
 - ☐ E) Tracheoesophageal fistula



38. A 27-year-old primigravid woman at 31 weeks' gestation is admitted to the hospital 60 minutes after awakening with painless bright red vaginal bleeding. Pregnancy had been uncomplicated. Her temperature is 37°C (98.6°F), pulse is 96/min, respirations are 22/min, and blood pressure is 118/84 mm Hg. The uterus is soft and nontender. The fundal height is 34 cm. The fetus is in an oblique lie with the head in the left lower quadrant. External fetal monitoring shows no contractions. The fetal heart rate is 156/min with no decelerations and normal variability. There is minimal flow of bright red blood at the introitus. Which of the following is the most appropriate next step in management?
- ☐ A) Induction of labor with oxytocin
 - ☐ B) Intravenous administration of fluids and observation
 - ☐ C) Intravenous administration of magnesium sulfate
 - ☐ D) Amnioinfusion
 - ☐ E) Immediate cesarean delivery



39. An otherwise asymptomatic 37-year-old woman, gravida 3, para 3, comes to the physician because of irregular vaginal bleeding since she underwent tubal ligation 2 years ago. Prior to tubal ligation, she used an oral contraceptive for 10 years. Since tubal ligation, menses have been occurring at 21- to 56-day intervals and lasting 5 to 13 days. All three of her deliveries were uncomplicated. She has no history of serious illness and takes no medications. She is 168 cm (5 ft 6 in) tall and weighs 90 kg (198 lb); BMI is 32 kg/m². Physical examination shows mild hirsutism over the chin and abdomen. Pelvic examination shows no abnormalities. An abnormality of which of the following serum concentrations is most consistent with the suspected diagnosis?
- ☐ A) Estradiol
 - ☐ B) Follicle-stimulating hormone
 - ☐ C) Prolactin
 - ☐ D) Testosterone
 - ☐ E) Thyroid-stimulating hormone



40. A previously healthy 32-year-old primigravid woman at 8 weeks' gestation comes to the emergency department 1 hour after awakening with vaginal bleeding and abdominal cramping. She has not had her first prenatal visit. Her pulse is 84/min, and blood pressure is 127/72 mm Hg. Pelvic examination shows mild bleeding from a closed cervical os. The uterus is consistent in size with an 8-week gestation. Endovaginal ultrasonography shows a gestational sac with an 8-week fetal pole and visible fetal cardiac activity. Which of the following is the most appropriate next step in management?
- ☐ A) Determination of blood type and Rh factor
 - ☐ B) Measurement of serum β -hCG concentration
 - ☐ C) Intramuscular medroxyprogesterone therapy
 - ☐ D) Oral progesterone therapy
 - ☐ E) Dilatation and curettage

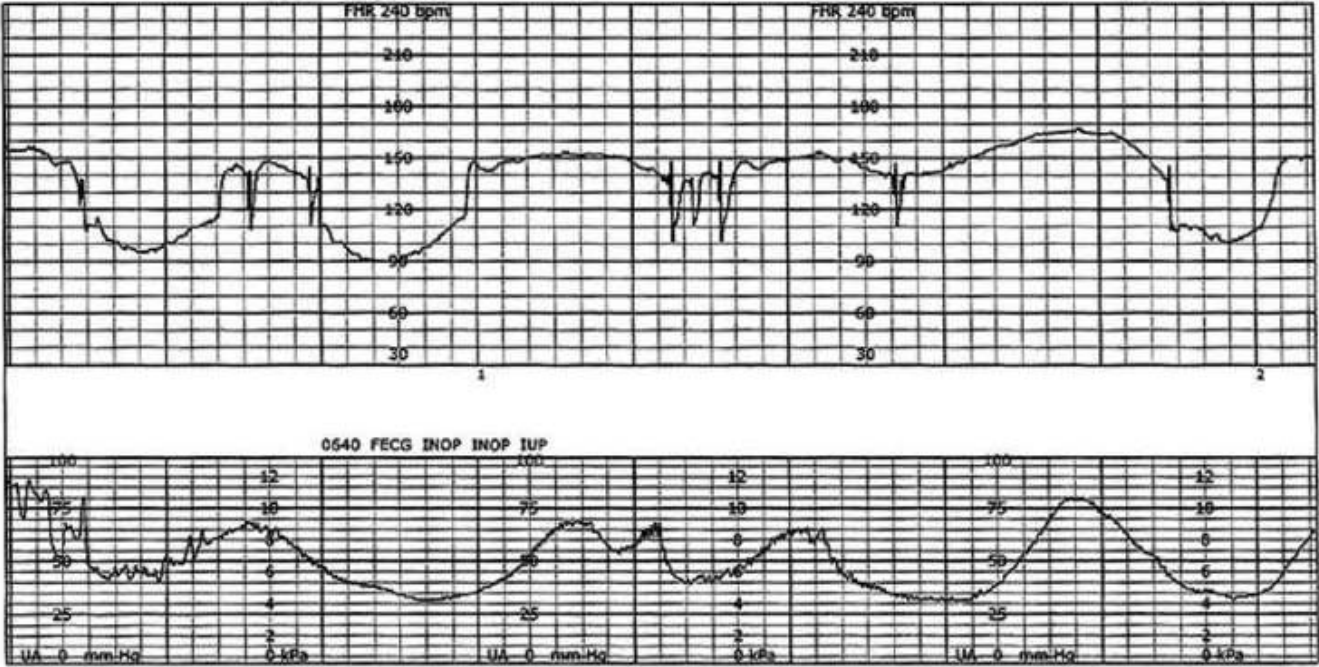


41. A previously healthy 27-year-old woman comes to the physician because of irregular menstrual periods during the past year. Menses now occur at 2- to 14-week intervals and last 5 to 12 days; her last menstrual period was 2 weeks ago and lasted 12 days. Menses previously occurred at regular 30-day intervals and lasted 5 days. She is currently sexually active with two partners and uses condoms consistently. She is 157 cm (5 ft 2 in) tall and weighs 104 kg (230 lb); BMI is 42 kg/m². Pelvic examination shows no abnormalities. This patient is at greatest risk for which of the following types of cancer?
- ☐ A) Breast
 - ☐ B) Colon
 - ☐ C) Endometrial
 - ☐ D) Ovarian
 - ☐ E) Pancreatic



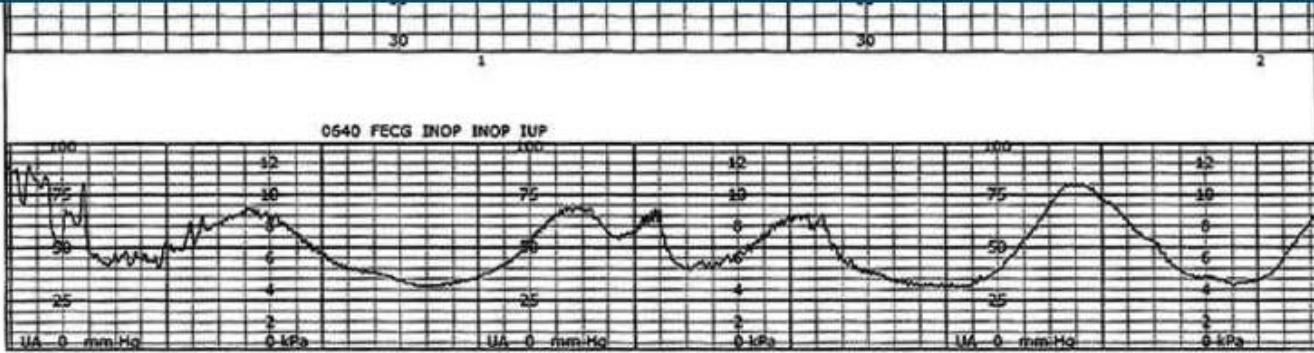
42. A 45-year-old woman comes to the physician for a follow-up examination after a routine mammography showed a 1-cm density in the upper outer quadrant of her left breast. On physical examination, a mass is not palpable. Ultrasonography shows a 1-cm hypoechoic mass in the upper outer quadrant of the left breast. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) MRI of the left breast
 - ☐ B) Needle-guided excisional biopsy
 - ☐ C) Repeat clinical examination in 3 months
 - ☐ D) Sentinel lymph node biopsy
 - ☐ E) Ultrasound-guided core-needle biopsy





43. A 29-year-old primigravid woman at 39 weeks' gestation is admitted to the hospital in labor 2 hours after the onset of contractions and spontaneous rupture of membranes. Contractions occur every 2 minutes and last 45 seconds. Her pregnancy has been uncomplicated. She has no history of serious illness, and her only medication is a prenatal vitamin. Her pulse is 110/min, respirations are 20/min, and blood pressure is 120/80 mm Hg. The cervix is 5 cm dilated and 100% effaced; the vertex is at -2 station. A fetal heart tracing is shown. Labor progresses during the next 3 hours. At delivery the newborn is limp and pale. Apgar scores are 3 and

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- ☐ A) Fetal metabolic acidosis
 - ☐ B) Fetal respiratory acidosis
 - ☐ C) Fetal respiratory alkalosis
 - ☐ D) Maternal lactic acidosis
 - ☐ E) Maternal respiratory alkalosis

44. A 32-year-old nulligravid woman comes to the physician for counseling prior to conception. She and her husband would like to conceive within the next year. Menses occur at regular 28-day intervals and last 5 days. The patient and her husband are both healthy, take no medications, and are in a stable marital relationship. Neither has used tobacco, alcohol, or illicit drugs. She has a strong family history of alcohol use disorder, and her father had Guillain-Barré syndrome. She is 170 cm (5 ft 7 in) tall and weighs 80 kg (176 lb); BMI is 28 kg/m². Her blood pressure is 122/84 mm Hg. A grade 1/6, systolic flow murmur is heard. Examination shows multiple diffuse nodularity of the breasts with no discrete masses. Pelvic examination shows no abnormalities. There is a large verruca plantaris on the left foot. Which of the following is the most appropriate advice for this patient before conception?
- ☐ A) Genetic counseling
 - ☐ B) Weight loss
 - ☐ C) Chemical dependency evaluation
 - ☐ D) Mammography
 - ☐ E) Folic acid supplementation
 - ☐ F) Removal of the verruca plantaris



Previous



Next

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Lab Values



Calculator



Review



Help



Pause

45. A previously healthy 72-year-old woman comes to the physician because of a 3-week history of vulvar itching and discomfort despite treatment with over-the-counter antifungal creams. She is sexually active with one male partner. Physical examination shows several 1-cm, small, firm left inguinal lymph nodes. Pelvic examination shows mild edema and excoriations over the labia majora and minora. There is a 1-cm, tender, raised lesion with an ulcerated center and erythematous edges at the base of the left labium majus. The cervix, vaginal mucosa, uterus, and adnexa are normal. There is a white discharge with no odor in the vaginal canal. Which of the following is the most likely diagnosis?
- ☐ A) Condylomata acuminata
 - ☐ B) Herpes simplex
 - ☐ C) Primary syphilis
 - ☐ D) Vulvar carcinoma
 - ☐ E) Vulvar dystrophy
 - ☐ F) Vulvar vestibulitis



46. An 11-year-old girl is brought to the physician because her mother is concerned that her daughter is maturing too early. She reports that her daughter has had breast and pubic and axillary hair growth during the past 6 months. She also had the onset of her first menstrual period 2 weeks ago. The mother adds that none of her daughter's friends have begun their menses. The patient is at the 75th percentile for height and 50th percentile for weight and BMI. Breast and pubic hair development is sexual maturity rating stage 4. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Precocious puberty
 - ☐ B) Premature adrenarche
 - ☐ C) Premature menarche
 - ☐ D) Premature thelarche
 - ☐ E) Normal development



47. A 27-year-old primigravid woman at 10 weeks' gestation comes to the physician for her first prenatal visit. She feels well, and her pregnancy has been uncomplicated. Examination shows a uterus consistent in size with a 10-week gestation. A Pap smear shows a high-grade squamous intraepithelial lesion. Which of the following is the most appropriate next step in management?
- ☐ A) Repeat Pap smear in 3 months
 - ☐ B) Repeat Pap smear 6 weeks post partum
 - ☐ C) Administer human papillomavirus vaccine
 - ☐ D) Perform colposcopy with biopsy if indicated
 - ☐ E) Perform cryotherapy



48. A 1900-g (4-lb 4-oz) newborn has generalized lymphadenopathy on examination. She was delivered at term to a 25-year-old primigravid woman. Pregnancy and delivery were uncomplicated. The head circumference is below the 5th percentile for gestational age. Her temperature is 37°C (98.6°F), pulse is 150/min, and respirations are 40/min. Funduscopic examination shows chorioretinitis. The liver edge is palpated 4 cm below the right costal margin, and the spleen tip is palpated 2 cm below the left costal margin. Serum studies show AST activity of 247 U/L and ALT activity of 196 U/L. A CT scan of the head shows scattered intracerebral calcifications. Which of the following intrauterine organisms is the most likely cause of these findings?
- ☐ A) Cytomegalovirus
 - ☐ B) Hepatitis B virus
 - ☐ C) Parvovirus B19
 - ☐ D) Rubella virus
 - ☐ E) *Treponema pallidum*



Previous



Next

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Lab Values



Calculator



Review



Help



Pause

49. A 52-year-old woman comes for a routine health maintenance examination. She has had occasional vaginal spotting over the past 2 months. She has also had hot flashes and vaginal dryness that has been relieved by black cohosh, a natural food supplement. She also takes ginseng root tea daily. Her last menstrual period was 2 years ago. Examination shows no abnormalities. In addition to a Pap smear and mammography, which of the following is the most appropriate next step in diagnosis?
- ☐ A) Measurement of serum estradiol concentration
 - ☐ B) Measurement of serum follicle-stimulating hormone concentration
 - ☐ C) Toxicology screening
 - ☐ D) Progesterone challenge test
 - ☐ E) Endometrial biopsy



50. A 16-year-old patient, gravida 1, para 1, has increasingly severe pelvic pain and feels cold 1 day after emergent cesarean delivery at term for nonreassuring fetal testing. She had spontaneous rupture of membranes at 3 AM. Prophylactic antibiotics were administered prior to the incision. The cesarean section was performed under general anesthesia and lasted 40 minutes. She delivered a 4423-g (9-b 12-oz) newborn at 11 PM. She is lying in bed under several blankets. Her temperature is 38.9°C (102°F), pulse is 129/min, respirations are 16/min, and blood pressure is 110/70 mm Hg. The uterus is tender to palpation. The incision site is clean and dry, and there is no erythema. Which of the following is the strongest predisposing factor for this patient's condition?

- A) Age of the patient
- B) Duration of cesarean delivery
- C) Genetic predisposition
- D) Exposure to general anesthesia
- E) Time from rupture of membranes to delivery
- F) Use of antibiotic prophylaxis